

THE GARDEN OF THE HESPERIDES

What is the footprint of immortality when the fruit is not eaten, but guarded forever?

STEP ONE

FUNCTIONAL UNIT

A divine orchard is modeled as a guarded service, not as a place on a map.

FUNCTIONAL UNIT

One annual guarded presentation of three golden apples from the Hesperidean orchard gate.

REFERENCE FLOW

Three 80 mm apples with a 10 μm gold skin and one year of allocated orchard support.

TIME & GEOGRAPHY

One annual cycle, reconstructed as a western Mediterranean sacred orchard.

INTENDED USE

Narrative engineering reconstruction for educational communication only.



Canonical subject

Gnarled tree + three golden apples + coiled Ladon.



The model measures the annual service of guarded divine fruit, not the whole myth.

STEP ONE

SYSTEM BOUNDARY

The boundary stops at the gate: no consumption, no immortality allocation and no divine credit.



INCLUDED

Gold skin, fruit body, allocated trellis, stone enclosure, compost, water, electricity and inbound freight.

EXCLUDED

Dragon metabolism, nymph labour, divine magic, land occupation and downstream eating are outside scope.

CUT-OFF RULE

Minor ritual objects below 1% are excluded unless they affect the visual design or main finding.

DATA STATUS

Documented myth + literature benchmarks + engineering assumptions + transparent proxies.



Every modeled flow receives an emission factor, an explicit zero or a declared exclusion.

STEP TWO

PHYSICAL RECONSTRUCTION

The impossible fruit is treated as ordinary apple geometry with an extraordinary surface.



10 μm
average gold skin

GEOMETRY

Three spherical apples, 80 mm diameter. Surface area: 0.0603 m².

GOLD MASS

Gold density 19,300 kg/m³ gives 0.0116 kg Au per functional unit.

LIFETIME

Tree and wall are allocated by annual service rather than one-off construction.



A very thin metallic surface is still enough to become the model hotspot.

STEP TWO

LIFE CYCLE INVENTORY

The inventory is deliberately small: nine flows, one annual service and no hidden magic.

Flow	Activity	Unit	Basis
Gold skin	0.0116 kg	kg	10 µm surface
Fruit body	0.750 kg	kg	3 apples
Compost	50.0 kg	kg	annual soil
Metal upkeep	5.0 kg	kg	tools + gate
Water	12.0 m³	m³	irrigation
Electricity	25.0 kWh	kWh	pump + light
Freight	30.0 t·km	t·km	0.05 t x 600 km

ALLOCATION

Wall and trellis are allocated by service life; no avoided burden is claimed.

PROXY CONTROL

DEFRA 2026 is used first; external sources are disclosed only where needed.

EXCLUDED MASS

Labour, magic and dragon food are stated outside boundary, not silently ignored.



The calculation chain is activity data x emission factor = contribution.

STEP TWO

MAKING THE APPLES

The production story is a quiet sequence of surface, harvest and protection.



1. GROW

Apple biomass is the fruit core, not the divine value.

2. GILD

Gold skin creates visibility and most emissions.

3. GUARD

Ladon is boundary logic, not a modeled fuel user.

4. PRESENT

The service ends at the orchard gate.



The myth becomes measurable when the ritual is converted into a physical service.

STEP TWO

ENERGY & LOGISTICS

The garden is remote, irrigated and guarded, but its ordinary flows remain minor.



25

kWh / year

12

m³ water

30

t·km freight

RATIONALE

Small pump, gate lighting, dry-season water and a one-off material delivery proxy.



Even a remote sacred garden is a low-energy system compared with a gold surface.

STEP THREE

EMISSION FACTORS

DEFRA 2026 carries the ordinary world; gold needs a declared external proxy.

ID	Flow	Factor	Source
S09	Primary gold	25,463 kg/kg	S&P Global
S05	Average metals	3.822 kg/kg	DEFRA 2026
S04	Compost	0.112 kg/kg	DEFRA 2026
S07	Electricity incl. WTT	0.184 kg/kWh	DEFRA 2026
S08	HGV freight incl. WTT	0.127 kg/t·km	DEFRA 2026
S06	Water supply	0.191 kg/m³	DEFRA 2026
S01	Food & drink	3.701 kg/kg	DEFRA 2026

PROXY DISCLOSURE

Gold factor is Scope 1+2 mine intensity, so downstream refining is likely understated.

SOURCE CONTROL

Stable source IDs are linked to the full workbook and proxy register.



The model separates DEFRA factors from external proxies before showing the result.

STEP FOUR

BASELINE RESULT

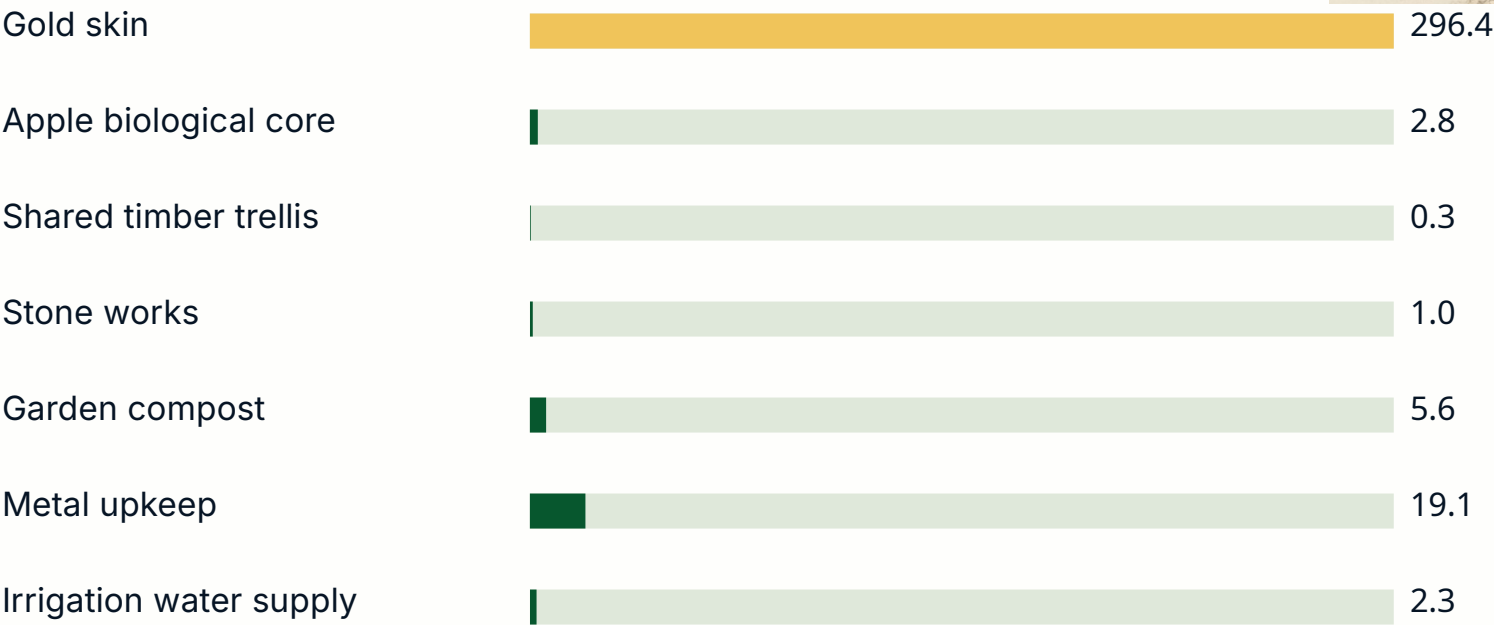
The garden is light; the golden skin is not.

336 kg

CO₂e per FU

HOTSPOT

Gold gilding layer: 296.4 kg CO₂e, 88.2% of the modeled total.



SHARE TEST

Minor flows are not enlarged visually. The bar length follows calculated values.



A visible gold surface controls the footprint more than water, freight and the wall combined.

STEP FIVE

SENSITIVITY A

Physical uncertainty: how thick is a mythic golden surface?

LOW PHYSICAL CASE
5 µm gold skin
0.0058 kg Au



BASELINE
10 µm gold skin
0.0116 kg Au



HEAVY GILDING
25 µm gold skin
0.0291 kg Au



RULE
Only the physical gold thickness changes. Mass, boundary, factors and minor flows remain fixed.

The baseline remains robust: gold thickness is the most consequential physical assumption.

STEP FIVE

SENSITIVITY B

Data uncertainty: the same apple changes when the gold source changes.



RULE

Only the gold emission factor changes. The same 10 µm skin and the same orchard inventory are retained.



Gold source selection changes the result by more than the entire ordinary garden.

STEP SIX

FINDING

What the model tells us.

01 SURFACE BEATS SETTING

The gold skin contributes 88.2% of the baseline footprint.

02 THE ORCHARD IS NOT THE HOTSPOT

Water, electricity and freight together remain below 4% of the total.

03 THICKNESS MATTERS

Changing the skin from 5 to 25 μm moves the result from 188 to 781 kg CO₂e.

04 PROXY HONESTY MATTERS

Gold source intensity shifts the same design from 139 to 484 kg CO₂e.



A thin golden surface can outweigh the whole garden.



The modeled insight survives without the myth: precious surface treatments dominate small support systems.

MYTHS & LEGENDS

Where impossible stories become traceable systems.



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